SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		

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Section / Batch:	AI&DS	**Date:**	10/10/26

Code of Conduct

I B.Muni Chandini (192524283) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student*: B.Muni Chandini*

Instructions

• Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.

• Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.

• Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.

• Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.

• Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Q Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	Q1
2. Intelligent Agents	Agent, percepts, actions, PEAS	Q1
3. Concept of Rationality	Performance measure, rational vs omniscient	Q1
4. Nature of Environments	Task-environment properties	Q1
5. Structure of Agents	Simple reflex and model-based reflex	Q1
6. Problem-Solving Agents	Goal/problem formulation, search, execution	Q2
7. Problem Formulation	States, actions, goal test, path cost; belief states	Q2, Q4
8. Searching for Solutions	Search graph, frontier, explored set	Q3
9. Uninformed Search Strategies	Error tracing of BFS, DFS, UCS	Q3
10. Search with Partial Information	Sensorless problems, belief states, percept update	Q4

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify correct / incorrect statements [Error Identification]

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"internal reasoning always follows the laws of logic" and "thinking rationally" – Turing Test is acting humanly, not thinking rationally.
S2	Incorrect	"occurred in the 1950s, before the 1956 Dartmouth workshop" – AI winters were later (1970s & late 1980s/90s). Dartmouth coined the term and launched the field.
S3	Incorrect	"always achieves the best possible outcome because it knows the actual result of every action in advance" – confuses rational with omniscient.
S4	Incorrect	Performance and Actuators are swapped. Performance = safe/fast/legal/comfortable trip; Actuators = steering, accelerator, brake, horn.
S5	Incorrect	"partially observable, stochastic, episodic" – Chess (no clock) is fully observable, deterministic, sequential, static, discrete.
S6	Correct	— (no error)

(b) Root cause analysis

Error	Root cause (misconception)
S1	Confusion among the four approaches to AI. The Turing Test is the classic example of "acting humanly" (behavioural indistinguishability). "Thinking rationally" refers to the laws-of-thought / logicist approach.
S2	Timeline of AI history inverted. Dartmouth (1956) marked the birth of AI as a field; the first AI winter followed the 1970s Lighthill report and funding cuts, the second after the expert-system boom of the 1980s.
S3	Equating rationality with omniscience. A rational agent maximises expected performance given its percept sequence and knowledge; it does not require perfect foresight of actual outcomes.
S4	Misassignment of PEAS components. Performance measure evaluates the agent's behaviour; actuators are the effectors the agent controls.
S5	Incorrect classification of environment properties. Chess has a fully visible board (fully observable), deterministic moves, sequential decisions, static (no change while deliberating), and discrete squares/pieces.

(c) Corrected statements with justification

St.	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine intelligent if a human interrogator cannot distinguish its behaviour from a human's. It is the standard example of the "acting humanly" approach.	Turing Test is behavioural (acting humanly), not internal logic (thinking rationally).
S2	The AI winters (periods of reduced funding) occurred in the 1970s and late 1980s/early 1990s, after the 1956 Dartmouth workshop that coined the term "artificial intelligence" and launched the field.	Historical sequence: Dartmouth first, winters later.

S3	A rational agent selects the action that maximises expected performance given its knowledge; it need not know actual outcomes in advance. A taxi that chose a route clear according to all available information was still rational even if a landslide later blocked it.	Rationality ≠ omniscience; expected performance is the criterion.
S4	PEAS of automated taxi – Performance: safe, fast, legal, comfortable trip. Environment: roads, traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	Performance evaluates outcome; actuators are the controls.
S5	Chess played without a clock is a fully observable, deterministic, sequential, static and discrete task environment.	Board fully visible, moves deterministic, decisions sequential, environment static, discrete state space.
S6	No change.	Correct description of a model-based reflex agent.

(d) Application to self-driving taxi case

(i) Four factors of rationality: (1) performance measure, (2) prior knowledge of environment, (3) actions the agent can perform, (4) percept sequence to date. The taxi chose a highway that every available traffic report showed clear. Given the information it had, the choice maximised expected performance → the choice was rational. The later landslide was unforeseeable, so the agent is not required to have predicted it.

(ii) Properties that make perfect foresight impossible: partially observable (cannot sense every future event), stochastic / nondeterministic (landslides, other drivers' behaviour), sequential and dynamic (world continues to change).

(iii) Simple-reflex taxi: reacts only to current percept (e.g., "if red light then stop"). It would behave poorly when the current percept is insufficient – e.g., approaching a junction where a pedestrian is hidden behind a truck. A model-based reflex taxi keeps an internal state (world model) updated by the transition model, allowing it to remember the hidden pedestrian or predicted traffic and act accordingly.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

Missionaries and Cannibals: 3M + 3C on left bank, boat capacity 1 or 2, cannot cross empty. Missionaries never outnumbered on either bank. Goal: everyone to right bank, fewest crossings.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M+1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	Agent order: search → goal formulation → problem formulation → execution
Note N2	State space contains 16 states, of which 10 are legal

(a) Correct / Incorrect identification

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	Missing boat location. Must include which bank the boat is on.
Initial state	Incorrect	Incomplete: boat side missing. Should be (3,3,Left) or (3,3,1).
Actions	Partially incorrect	Direction depends on boat side; empty boat forbidden is implied but actions must be conditional on boat location.
Goal test	Incorrect	M=C is not the goal. Goal is all on right bank: (0,0,Right).
Legality check	Incorrect	Only left bank checked. Both banks must satisfy the constraint (or have zero missionaries).
Path cost	Incorrect	Objective is fewest crossings → cost = number of boat trips (actions), not people carried.
Note N1	Incorrect	Order reversed. Correct: goal formulation → problem formulation → search → execution.
Note N2	Incorrect	With boat side: 4×4×2 = 32 possible states; typically ~11–16 are legal. Student's 16/10 omits boat and undercounts.

(b) Root cause analysis

Error	Root cause
State representation	Incomplete state abstraction – boat location is essential for determining legal moves and transitions.
Initial state / Goal test	Goal test must match the objective (everyone on right). M=C is an intermediate property, not the goal.
Legality check	Constraint must hold on both banks; checking only one side violates the problem's safety condition.
Path cost	Path cost must reflect the optimisation criterion stated in the problem (fewest crossings = unit cost per action).
Note N1	Standard problem-solving agent cycle is goal → formulate → search → execute; search cannot precede goal/problem formulation.
Note N2	State-space size calculation omitted the boat variable and did not enumerate legal states systematically.

(c) Corrected formulation

Component	Corrected formulation	Justification
State	(M_L, C_L, Boat) where Boat ∈ {L,R}	Boat side is required.
Initial	(3, 3, L)	All on left, boat on left.
Actions	Move k people (1 or 2, legal combinations) from boat's current side to the other.	Direction follows boat.
Goal test	(0, 0, R) – all on right bank.	Matches stated objective.
Legality	On each bank: if M>0 then M≥C (or M=0 is safe).	Both banks checked.

Path cost	1 per crossing (action).	Fewest crossings = unit cost.
N1	goal formulation → problem formulation → search → execution	Canonical agent cycle.
N2	~32 possible states; ~11–16 legal depending on exact counting.	Include boat variable.

(d) Application

(i) Boat loads from initial state (3,3,L):

Boat load	Resulting state (M,C,boat)	Legal? Reason
1M	(2, 3, R)	Illegal – left: 2M < 3C; right: 1M alone OK but left unsafe.
2M	(1, 3, R)	Illegal – left: 1M < 3C.
1C	(3, 2, R)	Legal – left 3M≥2C; right 0M safe.
2C	(3, 1, R)	Legal – left 3M≥1C; right 0M safe.
1M+1C	(2, 2, R)	Legal – left 2M=2C; right 1M=1C.

(ii) Example illegal state accepted by student's check: (1,0) on left (boat anywhere). Student's rule "C≤M or M=0" accepts it, but the right bank then has 2M and 3C → 2M < 3C, illegal.

(iii) Optimal solution cost = 11 crossings (standard minimal number of boat trips).

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

Graph edges: S–C:3, S–D:3, A–B:2, A–C:6, B–D:4, B–G:3, D–E:2, D–G:6, E–G:3. Start S, Goal G. Graph search, alpha expansion, goal test on removal, ties alpha.

(a) List of errors

No.	Where	What is wrong
1	BFS iter 3 – queue	After expanding D, queue should be [A, B, E, G] (A already present, new nodes appended). Student wrote [B, E, G, A] (wrong order).
2	BFS result path	Claimed path S–D–G. In BFS the first time G is dequeued the path is actually S–D–G only if parent pointers give that; but more critically the queue order error propagates. Correct shallowest path is S–C–A–B–G or via D, but cost claimed 9 is for S–D–G.
3	DFS iter 1 – stack	Neighbours of S are C, D (alpha). To pop alpha-first, push reverse → [C, D] top C. Student pushed [D, C] (top D) – wrong child order.
4	DFS iter 2	Because of wrong order, D expanded first; children of D (B,E,G alpha) should be pushed reverse so B pops first. Student's [G,E,B,C] has G on top – incorrect.
5	UCS – path & cost	UCS should find cheaper path S–D–E–G (cost 3+2+3=8). Student stopped at G:9 from D and never replaced with cheaper G via E.
6	UCS expansion of A	A is expanded with g=9 but A is not on any optimal path to G and after G is already reachable cheaper; also B:7 from D should allow B–G update.
7	Remarks R1, R2, R3	R1: explored set IS needed in graph search. R2: not all optimal (UCS should be 8; BFS/DFS not optimal). R3: fewest nodes ≠ best strategy in general.

(b) Root cause for each error

No.	Root cause (rule/concept violated)
1	BFS queue is FIFO; newly generated nodes are enqueued at the rear. Existing frontier nodes stay in place.
2–4	DFS child-ordering rule ("push so alpha-first is popped first") and graph-search non-reinsertion of explored/stack nodes were violated, producing wrong expansion order.
5–6	UCS must expand lowest-g first and replace a frontier entry when a cheaper path is found. Student never inserted/updated G via E (g=8) or via B (g=10).
7	Misunderstanding of optimality (only UCS is optimal with consistent costs), completeness, and the purpose of the explored set (prevents re-expansion in graphs with cycles/redundant paths).

(c) Corrected traces

(i) Corrected BFS (queue initially [S]):

Iter	Dequeued	Queue after Expansion	Explored
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[]	{…, G}

Order dequeued: S, C, D, A, B, E, G. Path recovered via parents (one possible shallowest): S–D–G (or S–C–A–B–G). Nodes removed: 7.

(ii) Corrected DFS (stack top on left; push reverse-alpha so alpha pops first):

Iter	Popped	Stack after Pushing Children
1	S	[C, D] (push D then C)
2	C	[A, D] (C's child A; D already on stack)
3	A	[B, D] (A's child B)
4	B	[G, D] (B's child G; D already present)
5	G (goal)	[D]

Path: S–C–A–B–G. Nodes removed: 5.

(iii) Corrected UCS (lowest g first; replace on cheaper path):

Iter	Expanded	g(n)	Priority Queue after Expansion
1	S	0	[C:3, D:3]
2	C	3	[D:3, A:9]
3	D	3	[E:5, B:7, A:9, G:9]
4	E	5	[B:7, G:8, A:9] (E→G cost 3 → g=8 replaces G:9)
5	B	7	[G:8, A:9] (B→G =10 >8, no replace)
6	G (goal)	8	[]

Path: S–D–E–G, cost 8. Nodes removed: 6.

(d) Application

(i) Comparison table:

Strategy	Path Found	Path Cost	Nodes Removed	Optimal?
BFS	S–D–G (or S–C–A–B–G)	9 (or 14)	7	No
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

(ii) Why explored set is needed (BFS iter 3 example): When D is expanded its neighbour A is already on the frontier (added when C was expanded). Without an explored/frontier check the algorithm would re-insert A, causing duplicate work and, in graphs containing cycles, possible non-termination. The explored set guarantees each node is expanded at most once.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Sensorless and percept-based belief-state search

(a) List of errors

No.	Where	What is wrong
1	Task A step 1 (Suck)	From {S1,S3,S5,S7} Suck on Left cleans Left → {S5,S7} is correct, but student later fails to reach a singleton goal.
2	Task A step 3 (Suck)	After Right the belief is {S6,S8}; Suck on Right cleans Right → should become {S8} only (S6 has Right Dirty → becomes S8). Student left {S6,S8}.
3	Task B step 0 update	Percept [Left, Dirty] retains only states with Agent=L and Left=D → {S1, S3}. Student wrote {S1,S2,S3,S6} (includes Right-side and Clean-Left states).
4	Task B step 1 prediction	From correct {S1,S3} Suck → {S5,S7}. Student's predicted set {S4,S5,S7,S8} is wrong (carried from previous error).
5	Task B step 2	Prediction and update cascade from earlier mistakes; final {S6} happens to be correct only by coincidence of later filtering, but intermediate sets are wrong.
6	Conclusion C1	Sensorless problems CAN be solved; a plan that works from every physical state in the belief (e.g. [Suck,Right,Suck] correctly executed) reaches a goal belief.
7	Conclusions C2 & C3	C2: goal not reached (Right still Dirty in S6). C3: number of belief states is 2^8=256, not 8.

(b) Root causes

No.	Root cause
1–2	Incorrect application of the deterministic transition on a belief set: Suck cleans the current square in every physical state, so states with that square Dirty become the corresponding Clean states.
3–5	Percept update must retain only physical states consistent with the percept. Student included states that contradict location or dirt status.
6–7	Misconception that uncertainty implies unsolvability; and confusion between physical states and the power-set of belief states.

(c) Corrected tables

(i) Task A – corrected belief states:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7} (Left cleaned)
2	Right	{S6, S8} (agent → Right)
3	Suck	{S8} (Right cleaned → only (R,C,C) remains – GOAL)

(ii) Task B – corrected prediction / update:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1..S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions:

C1 (corrected): The final belief state after the correctly executed plan is the singleton {S8}, which is a goal state. Sensorless problems are solvable by plans that succeed from every physical state in the initial belief.

C2 (corrected): The belief is the singleton {S6}, so the agent knows it is on the Right square with Left Clean and Right Dirty. The goal (both Clean) has NOT been reached; one more Suck is required.

(d) Application

(i) Number of possible belief states = 2^8 = 256 (power set of the 8 physical states). Belief-state search is harder because the state space is exponentially larger and each transition requires applying an action to every member of the current belief set.

(ii) Plan [Right, Suck] from {S1,S3,S5,S7}:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8} (Right cleaned)